

■ Java Internship Training Program

Week 1 — Fundamentals + OOP Deep Dive

Trainee: Avani Agarwal **Mentor:** Shivam **Duration:** 7 Days

■ Week 1 Goal

Strengthen Java syntax and understand OOP concepts practically. By the end of the week, the trainee should be able to independently write class hierarchies and build a small console-based application.

■ Day-Wise Training Plan

DAY 1

Topic: Java Setup, JDK/JVM/JRE concept, Data Types, Variables, Input/Output

Task: 10 basic programs: calculator, area of shapes, simple I/O operations

DAY 2

Topic: Conditionals (if/else/switch), Loops (for/while/do-while), break/continue

Task: Pattern printing (star pyramid) + Number problems: prime check, Fibonacci series

DAY 3

Topic: Arrays (1D & 2D), String methods, StringBuilder vs String

Task: Manual array sorting (bubble sort) + String reverse + Palindrome checker

DAY 4

Topic: OOP — Class, Object, Constructor (default + parameterized), this keyword

Task: Create Student class with name, rollNo, marks — add display() and getGrade() methods

DAY 5

Topic: Inheritance — extends, Method Overriding, super keyword, final keyword

Task: Animal → Dog, Cat, Bird hierarchy — each overrides sound() and move()

DAY 6

Topic: Polymorphism (compile-time & runtime), Abstraction, Abstract class

Task: Shape abstract class → Circle, Rectangle, Triangle — each implements area() and perimeter()

DAY 7

Topic: Interface, implementing multiple interfaces, interface vs abstract class

Task: Flyable + Swimmable interfaces → Duck class implements both

■ Week 1 Assignment

Build a **Student Management System (Console-based)** using Java with the following features:

- Add a new student (name, roll number, marks in 3 subjects)
- Display all students
- Search student by roll number
- Calculate average marks and display grade (A/B/C/D/F)
- Use proper OOP — Student class, separate Main class

Expected Time: 3–4 hours **Submit via:** GitHub

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Week 1 — Assessment: 20 Questions

Trainee Name: _____ Date: _____ Score: _____ / 20

SECTION A — Theory & Concepts (Q1–Q7) [1 mark each]

Q1. What is the difference between JVM, JDK, and JRE? Explain in one line.

Hint: JDK = Development Kit, JRE = Runtime, JVM = Virtual Machine

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

Q2. How many primitive data types are there in Java? Write their names.

Hint: 8 types — byte, short, int, long...

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

Q3. What is a Constructor? How is it different from a normal method? (2 differences)

Hint: Return type? Name?

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

Q4. What is the use of the 'super' keyword in Inheritance? Explain with an example.

Hint: Calling parent class constructor/method

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

Q5. What is the difference between Method Overriding and Method Overloading?

Hint: Override = same signature, Overload = different parameters

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

Q6. Write 3 differences between an Abstract class and an Interface.

Hint: Multiple inheritance, constructor, access modifiers

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

Q7. What are 3 use cases of the 'this' keyword in Java?

Hint: Current object reference, constructor chaining...

Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _Answer: _

SECTION B — Predict the Output (Q8–Q12) [1 mark each]

Write the output of the code given below. Do NOT use an IDE — think on your own first.

Q8.

```
int x = 5;
int y = 2;
System.out.println(x / y);
System.out.println(x % y);
```

Output: _____

Q9.

```
String s = "Hello";
s = s + " World";
System.out.println(s.length());
```

```
System.out.println(s.charAt(0));
```

Output: _____

Q10.

```
class Animal {  
    Animal() { System.out.println("Animal"); }  
}  
class Dog extends Animal {  
    Dog() { System.out.println("Dog"); }  
}  
// In main:  
Dog d = new Dog();
```

Output: _____

Q11.

```
int[] arr = {5, 3, 8, 1, 9};  
int max = arr[0];  
for(int i = 1; i < arr.length; i++) {  
    if(arr[i] > max) max = arr[i];  
}  
System.out.println(max);
```

Output: _____

Q12.

```
abstract class Shape {  
    abstract double area();  
    void display() {  
        System.out.println("Area: " + area());  
    }  
}  
class Circle extends Shape {  
    double r = 7;  
    double area() { return 3.14 * r * r; }  
}  
// In main:  
new Circle().display();
```

Output: _____

SECTION C — Coding Questions (Q13–Q17) [2 marks each]

Write the following programs in Java. Clean code + proper naming required.

Q13. Write a program that takes 5 numbers as input from the user and prints their sum, average, maximum, and minimum.

■ Arrays or simple variables — both are allowed.

Q14. Create a class 'Rectangle' with length and width fields. It should have a constructor, area() method, and perimeter() method. Create 3 different Rectangle objects in Main.

■ Initialize values via constructor; getters are optional.

Q15. Use Inheritance: Vehicle (parent) → Car, Bike (child). Vehicle should have an abstract method fuel(). Car and Bike should each print their own fuel type.

■ Use an abstract class, not an interface.

Q16. Write a program that reverses a string and checks whether it is a palindrome or not. Example: 'madam' → palindrome, 'hello' → not.

■ You can use StringBuilder or a loop.

Q17. Create an interface 'Printable' with a method print(). Classes 'Invoice' and 'Report' should both implement this interface and print their own message. In Main, call print() on both objects using a

Printable reference — demonstrate runtime polymorphism.

■ Use Printable reference → runtime polymorphism.

SECTION D — Scenario Based (Q18–Q20) [2 marks each]

Q18. You have a 'BankAccount' class. A junior developer has kept the balance field public — anyone can directly set `account.balance = -5000`. Why is this wrong? Write the corrected version using Encapsulation so that the balance cannot become negative.

■ Getter/Setter + validation in setter

Q19. You need to build a school system. Teacher and Student are both Person. Person has common fields name and age. Teacher has a subject field; Student has a rollNumber field. Both have an `introduce()` method that gives different output. Design and implement the class hierarchy.

■ Use Inheritance + Method Overriding

Q20. Interface 'Drawable' has a method `draw()`. Classes Circle, Rectangle, and Triangle all implement it. In Main, create a Drawable array containing all three objects and call `draw()` on each using a loop. Which OOP concept does this demonstrate? Explain.

■ Runtime Polymorphism — Drawable reference → different implementations

Section	Questions	Marks Each	Total
A — Theory	Q1 – Q7	1	7
B — Output Predict	Q8 – Q12	1	5
C — Coding	Q13 – Q17	2	10

D — Scenario Based	Q18 – Q20	2	6
		Grand Total	28